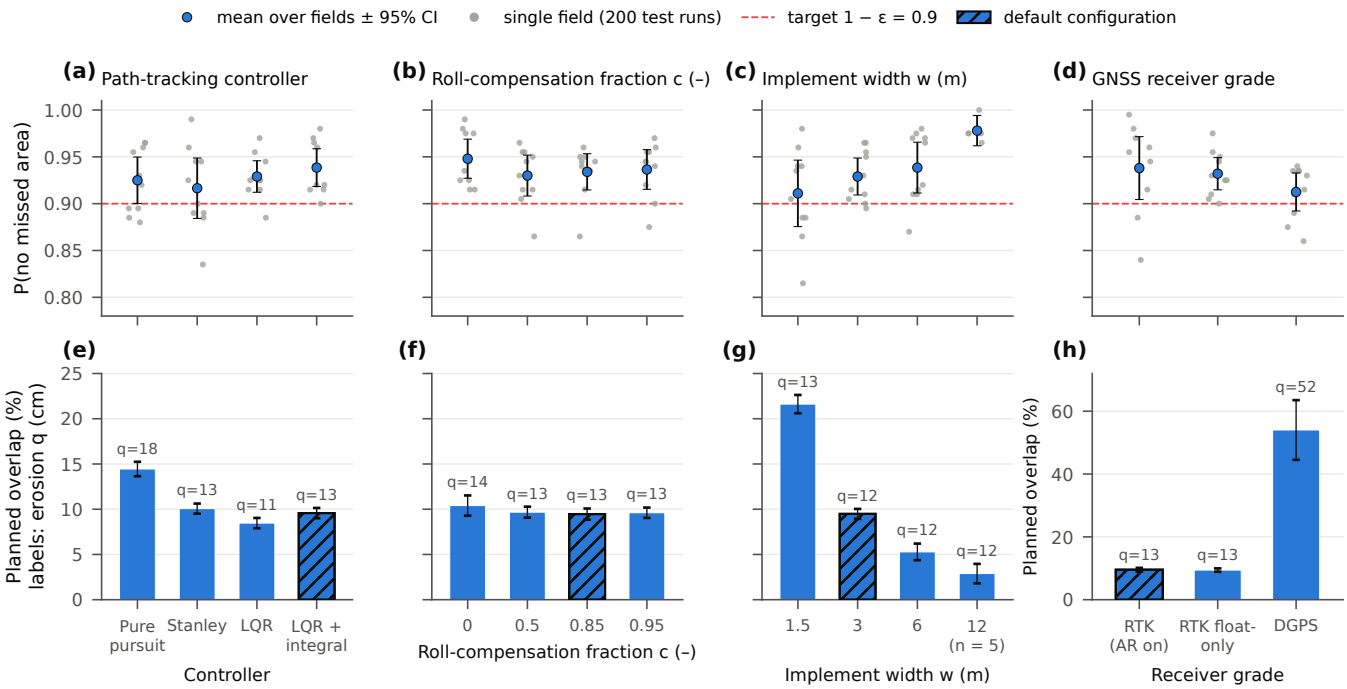
mean over fields ± 95% CI
single field (200 test runs)
target 1 − ε = 0.9
default configuration
(a) Path-tracking controller
(b) Roll-compensation fraction c (–)
(c) Implement width w (m)
(d) GNSS receiver grade
P(no missed area)
1.00
0.95
0.90
0.85
0.80
(e)
(f)
(g)
(h)
Planned overlap (%)
labels: erosion q (cm)
25
20
15
10
5
0
q=18
q=13
q=11
q=13
q=14
q=13
q=13
q=13
q=13
q=12
q=12
q=12
q=52
60
40
20
Pure pursuit
Stanley
LQR
LQR + integral
Controller
0
0.5
0.85
0.95
Roll-compensation fraction c (–)
1.5
3
6
12 (n = 5)
Implement width w (m)
RTK (AR on)
RTK float-only
DGPS
Receiver grade